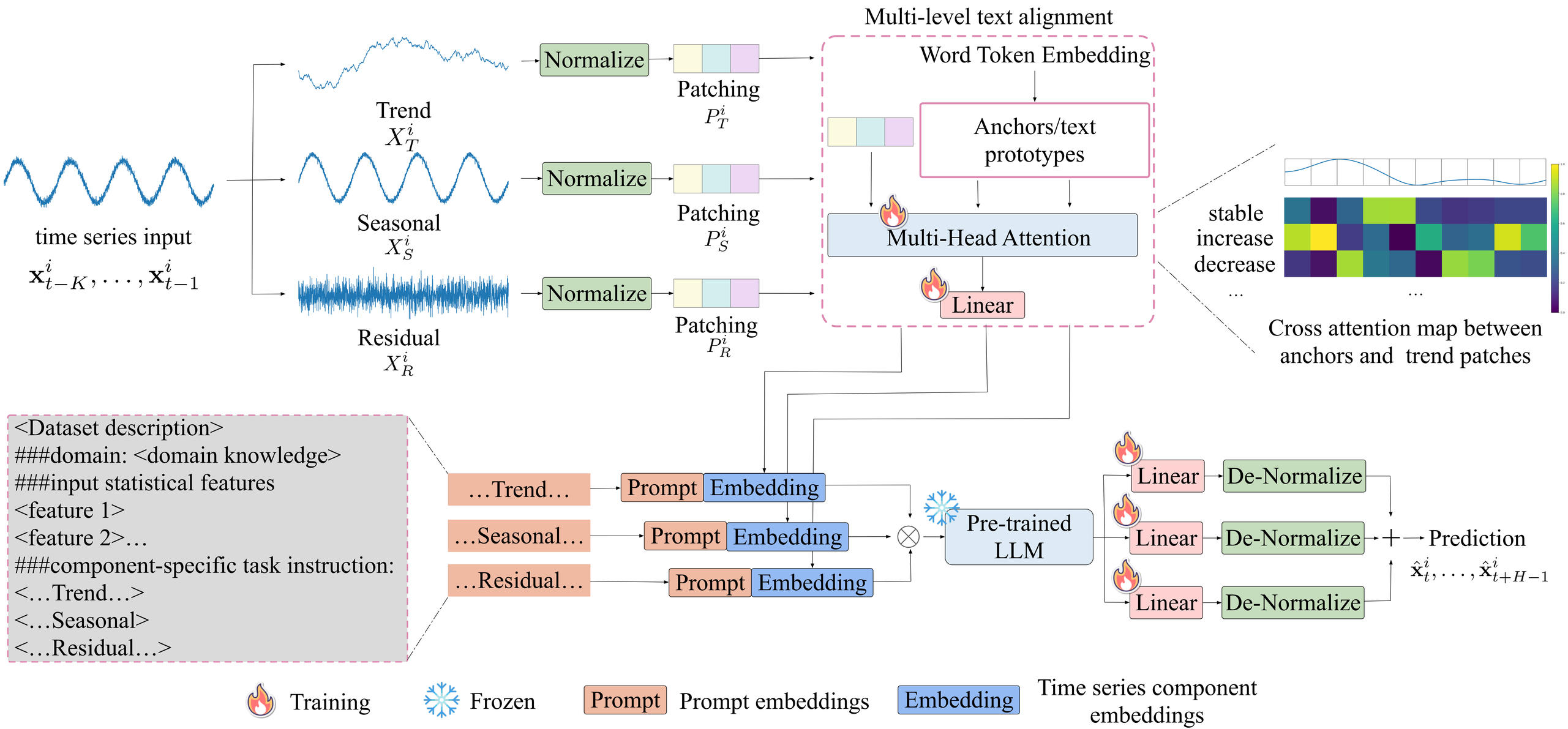
Multi-level text alignment
Word Token Embedding
Anchors/text prototypes
Multi-Head Attention
Linear
Normalize
Normalize
Normalize
Patching $P_T^i$
Patching $P_S^i$
Patching $P_R^i$
Trend $X_T^i$
Seasonal $X_S^i$
Residual $X_R^i$
time series input
$\mathbf{x}_{t-K}^i, \ldots, \mathbf{x}_{t-1}^i$
stable
increase
decrease
...
...
Cross attention map between anchors and trend patches
<Dataset description>
###domain: <domain knowledge>
###input statistical features
<feature 1>
<feature 2>...
###component-specific task instruction:
<...Trend...>
<...Seasonal>
<...Residual...>
...Trend...
...Seasonal...
...Residual...
Prompt
Embedding
Pre-trained LLM
Linear
De-Normalize
Prediction
$\hat{\mathbf{x}}_t^i, \ldots, \hat{\mathbf{x}}_{t+H-1}^i$
Training
Frozen
Prompt
Prompt embeddings
Embedding
Time series component embeddings